APH205 Final Report

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1. **Critique**

1.1 Potential limitations

1.1.1 Data and sample collected

The sample is underrepresented, this study only selected students from a public junior high school in a suburban area of Jiangsu Province, China. The local area and cultural background are homogeneous, and there was no comparison between adolescents' psychological adaptation abilities across regions with different levels of economic development (rural versus urban). This makes it difficult to generalize the findings to other regions in China and adolescents in other countries. Additionally,excluding third-year junior high school students might lead to results that bias towards the psychological characteristics of younger students, overlooking the unique pressures faced by older students (such as college entrance anxiety)

Data collection is subjective**.** This study only used self-rated scales (PHQ-9, GAD-7, ISI) to assess psychological symptoms without objective indicators, this may lead to social desirability bias or recall bias. Moreover, the pandemic impact scores focus solely on negative effects and do not include positive impact options, resulting in one-sided bias in data collection

The data is delayed. The data were analyzed two years later. During the epidemic period, policies, social mentality and psychological adaptation strategies of adolescents may have changed, which limited the timeliness of the conclusions.

1.1.2 The limitations of the analysis method.

The study only uses LLCA to analysis and does not try other methods to verify the reliability of the results. Moreover, although the study includes variables such as gender, grade, BMI, etc., it does not control key confounding factors (such as behavioral factors), which may overestimate the independence of the impact of the epidemic.

1.2 Alternative Approaches and Improvement Suggestions

1.2.1 Sample and Data

Use Multi-center Sampling Include adolescents from diverse regions schools and socioeconomic backgrounds to enhance sample diversity. Adding physiological metrics, use wearable devices to collect objective data (e.g., sleep quality, heart rate variability) to reduce self-report biases.

Incorporate variables such as social support, physical exercise frequency, media exposure, and pre-existing mental health conditions to build a comprehensive predictive model.

Add control groups: Compare students who experienced prolonged COVID-19 lockdowns with those who did not to isolate pandemic-specific effects.

1.2.2 Statistic method

For the multiple logistic regression model, it is suggested to test the multicollinearity and calculate the variance inflation factor to predict the collinearity between variables.

It is recommended to add machine learning methods if the data volume is sufficient, try to identify complex interactions (such as the synergistic effect of gender and academic stress

Bayesian network analysis was used to explore the causal relationship between variables and enhance mechanism interpretation.

2. Data analysis

In this section, i will use SAS to analysis the provided database “APH205 Final Report Database”, aim to find the relationship between perceived epidemic impacts recorded at the baseline and mental symptoms at the 4th follow-up.

The dependent variable is PHQ9_score_F4, GAD7_score_F4, ISI_score_F4, independent variable isCOVID19_impact_score, COVID19_impact_studypractice, COVID19_impact_familyincome, COVID19_impact_familyhealth, COVID19_impact_relationship, COVID19_impact_dailylife.

2.1 Data preparetion

First, conduct descriptive statistical analysis on the original dataset and identify missing values in the variables. For continuous variables (Height_BL, Weight_BL, PHQ9_score_BL, GAD7_score_BL, ISI_score_BL), apply mean to address missing values. For categorical variables (PHQ9_cat_BL, GAD7_cat_BL, ISI_cat_BL), use mode. The final dataset contains 1,192 complete observations.

2.2 Normality analysis

Before the correlation analysis, it is necessary to determine whether the independent variables are normally distributed. The results of all Shapiro-Wilk test and Kolmogorov-Smirnov test are significant, indicating that they do not conform to normal distribution (table 1).

	Shapiro-Wilk test	Kolmogorov-Smirnov test
COVID19_impact_score	0.914598, p<0.0001	0.112549, p<0.0100
COVID19_impact_studypractice	38.34978, p<0.0001	413, p<0.0001
COVID19_impact_familyincome	35.47266, p<0.0001	376.5, p<0.0001
COVID19_impact_familyhealth	14.57894, p<0.0001	114, p<0.0001
COVID19_impact_relationship	14.16044, p<0.0001	113, p<0,0001
COVID19_impact_dailylife	0.829097, p<0.0001	0.237106, p<0.0001

Table 1

2.3 Correlation analysis

According to the results, all the correlation coefficients between the dependent variables and independent variables were greater than zero, indicating that COVID-19perceived epidemic impacts would lead to higher PHQ9, GAD7, and ISI scores at the fourth follow-up, that is, mental symptoms worsened. However, it is worth noting that the correlation between COVID19_impact_dailylife and ISI scores was not very significant (Figure 1).

Spearman Correlation Coefficients, N = 1192
Prob > |r| under H0: Rho=0

	COVID19_impact_score	COVID19_impact_studypractice	COVID19_impact_familyincome	COVID19_impact_familyhealth	COVID19_impact_relationship	COVID19_impact_dailylife
PHQ9_score_F4 PHQ9_score_F4	0.19509 <.0001	0.19757 <.0001	0.12716 <.0001	0.08429 0.0036	0.16829 <.0001	0.12378 <.0001
GAD7_score_F4 GAD7_score_F4	0.20145 <.0001	0.20967 <.0001	0.12171 <.0001	0.06021 0.0377	0.17475 <.0001	0.13402 <.0001
ISI_score_F4 ISI_score_F4	0.16051 <.0001	0.14947 <.0001	0.11483 <.0001	0.05122 0.0771	0.13830 <.0001	0.12552 <.0001

Figure 1

2.4 Regression analysis

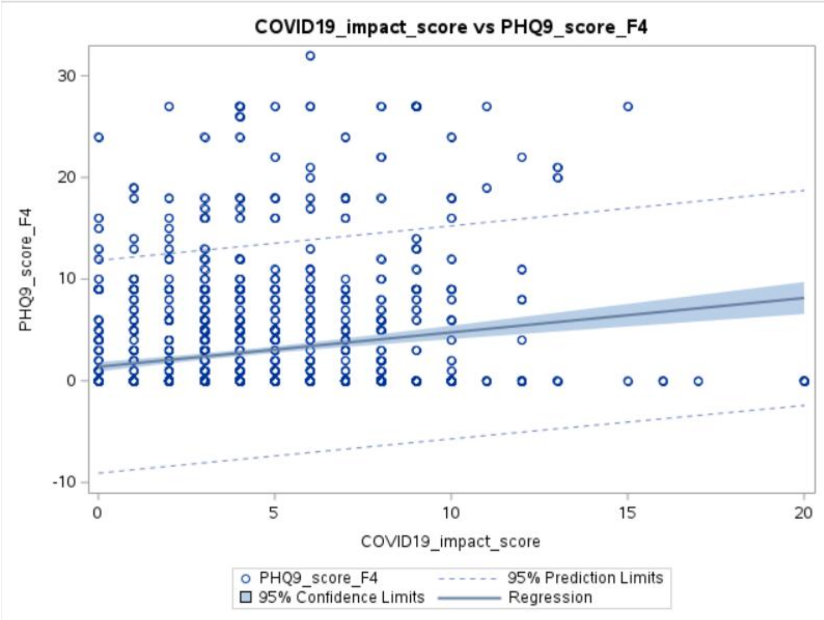
First, a multiple regression analysis was established. However, since multicollinearity was detected among the independent variables, subsequent regression analyses retained only COVID19_impact_score as the independent variable. To account for confounding factors, Age_BL, Sex, Ethnicity, Height_BL, and Weight_BL were included as covariates in the model.

COVID19_impact_score and PHQ9_score_F4：

The ANOVA results indicated that the overall model was statistically significant (F-value = 11.48, $p < 0.001$).

The t-test revealed that COVID19_impact_score had a significant effect on PHQ9 scores (T-value = 6.86, $p < 0.001$).

Thus, COVID19_impact_score have a statistically significant influence on PHQ9 scores.



COVID19_impact_score and GAD7_score_F4:

The ANOVA results indicated that the overall model was statistically significant (F-value = 43.43, $p < 0.001$).

The t-test revealed that COVID19_impact_score had a no significant effect on GAD7 scores (T-value =1.15, p=0.0514).

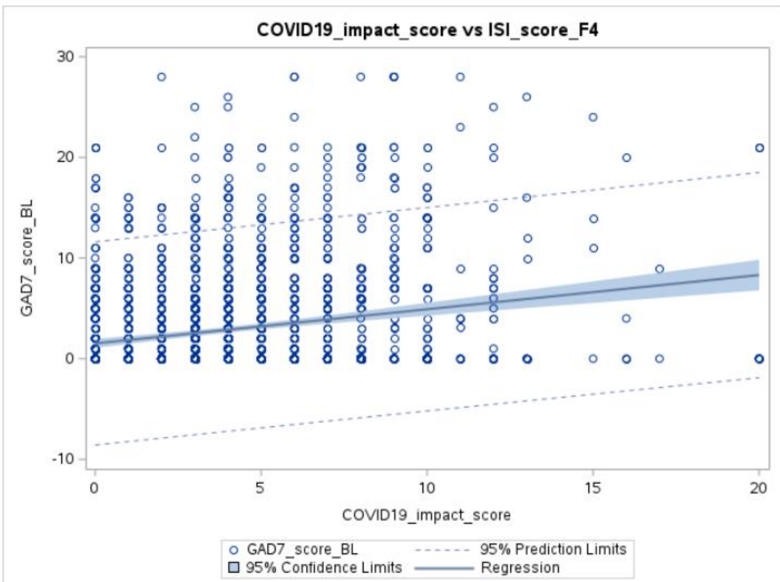
Thus, COVID19_impact_score have no statistically significant influence on GAD7 scores.

COVID19_impact_score and ISI_score_F4:

The ANOVA results indicated that the overall model was statistically significant (F-value = 40.22, p < 0.001).

The t-test revealed that COVID19_impact_score had a no significant effect on ISI scores (T-value =2.14, p=0.0322).

Thus, COVID19_impact_score have no statistically significant influence on ISI scores.



In general, COVID19_impact_score had a significant effect on ISI and GAD7 scores at the fourth follow-up. The higher the score of COVID19_impact_score, the worse the mental state of students at the fourth follow-up.

2.5 Logistic regression

Logistic regression was performed on the categorical variables PHQ9_cat_F4, GAD7_cat_F4, ISI_cat_F4 and COVID19_impact_score. In Figure 2, the OR values of ISI, GAD7 and PHQ9 were all greater than 1, and the confidence interval did not cross 1. This indicates that higher COVID-19 impact scores are significantly positively correlated with the risk of insomnia, anxiety and depression symptoms.

The OR value of PHQ9 may be the highest, further supporting the strong effect of COVID-19 impact score on depression symptoms. The confidence intervals of all results were narrow and far from 1, indicating that the association strength was statistically significant (p<0.001), which was consistent with the results of ANOVA and T-test.

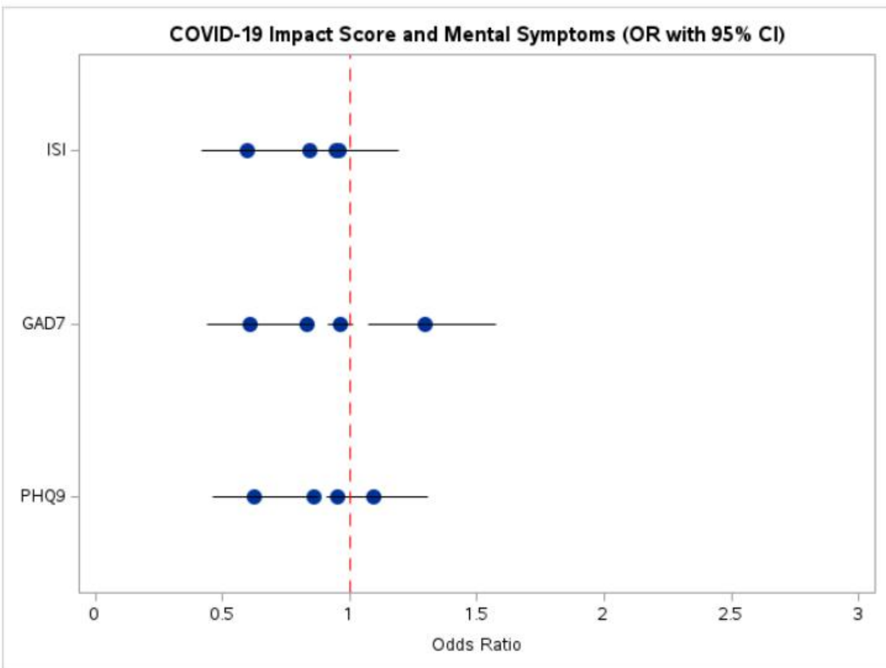


Figure 2

3. Conclusion

In general, COVID-19 significantly affect students mental health, especially in terms of depressive symptoms and insomnia. School should pay attention to the psychological state of students after the lifting of the lockdown。

Word count: 999 (unincluded table and figures).

Reference list:

R Rong, Q Xu, KP Jordan, Y Chen. Perceived Epidemic Impacts and Mental Symptom Trajectories in Adolescents Back to School After COVID-19 Restriction: A Longitudinal Latent Class Analysis. J Adolesc Health. 2024; 74: 487-495.

APEENDIX:

SAS code:

```
libname final "/home/u64011409/APH205final";
run;
proc print data=final.data;
run;
proc means data=final.data n nmiss;
run;
/*Fill in the data*/
/*Weight bl*/
proc means data=final.data mean;
  var Weight_BL;
  output out=mean_weight(drop=_type_ _freq_) mean=mean_weight;
run;
proc sql noprint;
  select mean_weight into :mean_weight trimmed
  from mean_weight;
quit;
data data_filled;
  set final.data;
  if missing(Weight_BL) then Weight_BL = &mean_weight;
run;
proc means data=data_filled n nmiss;
run;

/*hight bl*/
proc means data=final.data mean;
  var Height_BL;
  output out=mean_height(drop=_type_ _freq_) mean=mean_height;
run;
proc sql noprint;
  select mean_height into :mean_height trimmed
  from mean_height;
quit;
data data_filled1;
  set data_filled;
  if missing(Height_BL) then Height_BL = &mean_height;
run;
```

```
proc means data=data_filled1 n nmiss;
run;

/*PHQ SCORE BL*/
proc means data=final.data mean;
  var PHQ9_score_BL;
  output out=mean_PHQ9_score_BL(drop=_type_ _freq_) mean=mean_PHQ9_score_BL;
run;
proc sql noprint;
  select mean_PHQ9_score_BL into :mean_PHQ9_score_BL trimmed
  from mean_PHQ9_score_BL;
quit;
data data_filled2;
  set data_filled1;
  if missing(PHQ9_score_BL) then PHQ9_score_BL = &mean_PHQ9_score_BL;
run;
proc means data=data_filled2 n nmiss;
run;

/*GAD7 SCORE BL*/
proc means data=final.data mean;
  var GAD7_score_BL;
  output out=mean_GAD7_score_BL(drop=_type_ _freq_) mean=mean_GAD7_score_BL;
run;
proc sql noprint;
  select mean_GAD7_score_BL into :mean_GAD7_score_BL trimmed
  from mean_GAD7_score_BL;
quit;
data data_filled3;
  set data_filled2;
  if missing(GAD7_score_BL) then GAD7_score_BL = &mean_GAD7_score_BL;
run;
proc means data=data_filled3 n nmiss;
run;

/*ISI SCORE BL*/
proc means data=final.data mean;
  var ISI_score_BL;
  output out=mean_ISI_score_BL(drop=_type_ _freq_) mean=mean_ISI_score_BL;
run;
proc sql noprint;
  select mean_ISI_score_BL into :mean_ISI_score_BL trimmed
  from mean_ISI_score_BL;
quit;
data data_filled4;
  set data_filled3;
  if missing(ISI_score_BL) then ISI_score_BL = &mean_ISI_score_BL;
run;
proc means data=data_filled4 n nmiss;
run;

/*PHQ CAT BL*/
proc freq data=final.data noprint;
  tables PHQ9_cat_BL / out=mode_phq9;
run;
proc sql noprint;
  select PHQ9_cat_BL into :mode_value
  from mode_phq9
  having count=max(count);
quit;
data data_filled5;
  set data_filled4;
  if missing(PHQ9_cat_BL) then PHQ9_cat_BL = &mode_value;
run;
proc means data=data_filled5 n nmiss;
run;

/*GAD7_cat_BL*/
proc freq data=final.data noprint;
  tables GAD7_cat_BL / out=mode_gad7;
run;
proc sql noprint;
  select GAD7_cat_BL into :mode_value
  from mode_gad7
```

```
  having count=max(count);
quit;
data data_filled6;
  set data_filled5;
  if missing(GAD7_cat_BL) then GAD7_cat_BL = &mode_value;
run;
proc means data=data_filled6 n nmiss;
run;

/*ISI_cat_BL*/
proc freq data=final.data noprint;
  tables ISI_cat_BL / out=mode_isi;
run;
proc sql noprint;
  select ISI_cat_BL into :mode_value
  from mode_isi
  having count=max(count);
quit;
data analysis_data;
  set data_filled6;
  if missing(ISI_cat_BL) then ISI_cat_BL = &mode_value;
run;
proc means data=analysis_data n nmiss;
run;

/*Discription statitic*/
proc means data=analysis_data n nmiss mean std min max;
run;
/*normality test*/
proc univariate data=analysis_data normal;  /* normal选项执行正态性检验 */
  var COVID19_impact_score
        COVID19_impact_studypractice
        COVID19_impact_familyincome
        COVID19_impact_familyhealth
        COVID19_impact_relationship
        COVID19_impact_dailylife;
  histogram / normal;
```

```
  qqplot / normal(mu=est sigma=est);
run;
/*Correlation test*/
proc corr data=analysis_data spearman;
  var   COVID19_impact_score
        COVID19_impact_studypractice
        COVID19_impact_familyincome
        COVID19_impact_familyhealth
        COVID19_impact_relationship
        COVID19_impact_dailylife;
  with PHQ9_score_F4
        GAD7_score_F4
        ISI_score_F4 ;
run;
/*mutilinear anlysis*/
proc reg data=analysis_data plots=fit(nolimits);
  model PHQ9_score_F4 = COVID19_impact_score
        COVID19_impact_studypractice
        COVID19_impact_familyincome
        COVID19_impact_familyhealth
        COVID19_impact_relationship
        COVID19_impact_dailylife;
  output out=reg_out predicted=pred residual=resid;
run;

/*Linear regerssion*/
proc reg data=analysis_data plots=fit(nolimits);
  model PHQ9_score_F4 =COVID19_impact_score Age_BL Sex Ethnicity Height_BL Weight_BL;
  output out=reg_out predicted=pred residual=resid;
run;

proc reg data=analysis_data plots=fit(nolimits);
  model GAD7_score_F4 = COVID19_impact_score GAD7_score_BL Age_BL Sex Ethnicity Height_BL Weight_BL;
  output out=reg_out predicted=pred residual=resid;
run;

proc reg data=analysis_data plots=fit(nolimits);
```

```
  model ISI_score_F4 = COVID19_impact_score ISI_score_BL Age_BL Sex Ethnicity Height_BL Weight_BL;
  output out=reg_out predicted=pred residual=resid;
run;

/*plot*/
proc sgplot data=analysis_data;
  scatter x=COVID19_impact_score  y=PHQ9_score_F4;
  reg x=COVID19_impact_score  y=PHQ9_score_F4 / clm cli;
  title "COVID19_impact_score vs PHQ9_score_F4";
run;

proc sgplot data=analysis_data;
  scatter x=COVID19_impact_score y=GAD7_score_BL;
  reg x=COVID19_impact_score y=GAD7_score_F4 / clm cli;
  title "COVID19_impact_score vs GAD7_score_F4";
run;

proc sgplot data=analysis_data;
  scatter x=COVID19_impact_score y=GAD7_score_BL;
  reg x=COVID19_impact_score y=ISI_score_F4 / clm cli;
  title "COVID19_impact_score vs ISI_score_F4";
run;

/*logistic regression*/
/* PHQ9 */
proc logistic data=analysis_data;
  model PHQ9_cat_F4 = COVID19_Impact_score PHQ9_score_BL Age_BL Sex / link=logit;
  oddsratio COVID19_Impact_score / cl=wald;
  ods output OddsRatios=OR_PHQ9;
run;

/* GAD7 */
proc logistic data=analysis_data;
  model GAD7_cat_F4 = COVID19_Impact_score GAD7_score_BL Age_BL Sex / link=logit;
  oddsratio COVID19_Impact_score / cl=wald;
  ods output OddsRatios=OR_GAD7;
run;

/* ISI */
proc logistic data=analysis_data;
  model ISI_cat_F4 = COVID19_Impact_score ISI_score_BL Age_BL Sex / link=logit;
  oddsratio COVID19_Impact_score / cl=wald;
  ods output OddsRatios=OR_ISI;
run;

data OR_combined;
  set OR_PHQ9(in=A) OR_GAD7(in=B) OR_ISI(in=C);
  if A then Outcome = "PHQ9";
  if B then Outcome = "GAD7";
  if C then Outcome = "ISI";
  rename LowerCL=LowerCL UpperCL=UpperCL;
run;
proc sgplot data=OR_combined noautolegend;
  title "COVID-19 Impact Score and Mental Symptoms (OR with 95% CI)";
  scatter y=Outcome x=OddsRatioEst / markerattrs=(symbol=circlefilled size=12);
  highlow y=Outcome low=LowerCL high=UpperCL / lineattrs=(color=black);
  refline 1 / axis=x lineattrs=(color=red pattern=dash);
  xaxis label="Odds Ratio" values=(0 to 3 by 0.5);
  yaxis display=(nolabel) discreteorder=data;
run;
```